

Geographical Customer Segmentation for Targeted Marketing

This project focuses on leveraging geographical customer segmentation to enhance targeted marketing efforts. By analyzing customer data and identifying regional patterns, the goal is to optimize marketing strategies, improve customer engagement, and drive sales. The project utilizes clustering techniques to segment customers based on their geographical locations and purchasing behaviors, enabling stakeholders to tailor marketing campaigns to specific regions and demographics.

2. Introduction

2.1 Background

In today's competitive market, businesses must ensure their marketing efforts are directed toward the right audience. Geographical segmentation allows companies to divide their customer base into regions with similar characteristics, enabling more personalized and effective marketing strategies.

2.2 Problem Statement

Marketing teams often struggle to identify which regions or demographics to target for maximum ROI. Without proper segmentation, marketing efforts may be wasted on irrelevant audiences, leading to decreased engagement and sales.

2.3 Project Goal

The goal of this project is to perform geographical customer segmentation using clustering algorithms to identify distinct customer groups based on location and behavior. These insights will help stakeholders design targeted marketing campaigns that resonate with specific regions and demographics.

3. Data Summary

The dataset used in this project was 6 months of sales history that includes:

- Customer demographics (age, gender, income, etc.)
- Geographical data (city, state, country, zip code)

- Purchase history (transaction amounts, frequency, product categories)

3.1 Data Description

The dataset contains 8153 rows and 27 columns, with features such as:

- Customer_ID: Unique identifier for each customer
- Location: Geographical information (city, state, zip code)
- Purchase_Amount: Total amount spent by the customer
- Age, Gender, Income: Demographic information

3.2 Data Challenges

- Missing values in demographic fields
 - Inconsistent formatting of geographical data
 - Outliers in purchase amounts
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4. Exploratory Data Analysis (EDA)

- Visualized data distribution using histograms, boxplots, and bar charts.
- Analyzed correlations between features using heatmaps
- Plotted and identified correlations between geographical location and purchase behavior.
- Handled missing values as a result of merging datasets. This only made up 2% of the entire dataset so they were removed.

5. Data Preprocessing

- **Geographical Data Cleaning:** Standardized city and state names, and extracted latitude and longitude for mapping.
- **Feature Engineering:** Created new features such as an engagement score, distance metrics, and binning zip codes by frequency.
- **Data Normalization/Scaling:** Scaled numeric features using StandardScaler, and OneHotEncoder for categorical features.
- **Algorithm Selection:** Used **K-Means Clustering** to segment customers based on geographical and behavioral features. Determined the optimal number of clusters using the Elbow Method.
- **Model Evaluation:** Evaluated the quality of clusters using KMeans. Used PCA for better visualization.

6. Visualization

- Created interactive maps to display customer segments.
 - Generated bar charts and heat maps to highlight regional purchasing trends.
 - Used Principal Component Analysis (PCA) to reduce dimensionality and better visualize.
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Results

Customer Segments

The clustering algorithm identified 3 distinct customer segments:

Cluster 0: High-demand areas with significant interest and competition.

Characteristics: located in diverse geographical areas, high number of bids and views

Cluster 1: Mid-range auction participation, balanced bidding behavior (not aggressive, not passive), more price-conscious buyers.

Cluster 2: Budget-conscious segment with low competition. Characteristics: less frequent bidding, lower sale price tolerance, likely to be new or cautious buyers.

Key Insights & Recommendations

- **Cluster 0** makes up **39.77%** of total revenue, with Cluster 1 contributing to revenue second most.
 - **Cluster 0 (High-Engagement, High-Bids, High Sale Price)** → Exclusive Auctions & VIP Access, Personalized Recommendations, Loyalty Perks, Urgency & FOMO.
 - **Cluster 1 (Moderate demand, Mid-price)** → Price-Match & Discount Alerts, Auction Bundles, Social Proof & Reviews, Segmented Email Campaigns.
 - **Cluster 2 (Low-interest, Low-price)** → First-Time Buyer Incentives, Educational Content, Referral Bonuses, Flash Sales & Clearance Events.
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This project successfully demonstrated the value of geographical customer segmentation for targeted marketing. By identifying distinct customer groups and their preferences, stakeholders can allocate resources more effectively and design campaigns that resonate with specific regions. Future work could include incorporating additional data sources, such as online behavior, to further refine segmentation.

The complete code and documentation for this project can be found on GitHub:
[Capstone Three Repository](#)
